

CKD Prediction: Feature Removal & Re-Evaluation Report

1. Executive Summary

Following the discovery that three urine test markers (Urine Albumin, Urine Protein, and Albumin-to-Creatinine Ratio) were deterministic rules directly dictating the CKD target label, these features were removed from the training pipeline. The dataset now uses the remaining 32 diverse clinical, demographic, and blood test features to prevent data leakage and provide a robust prediction model.

2. Retraining & Evaluation Strategy

The model training pipeline was updated to include all remaining 32 features. Stratified 5-Fold Cross Validation was performed for Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting models. The Streamlit UI was updated to remove the target-leaking urine markers and introduce the advanced clinical markers seamlessly.

3. Final Model Configuration

The new best-performing model selected for production is: LogisticRegression. It utilizes exactly 32 predictive features (excluding the leaked urine metrics).